

MSK IMCQ-6

KEY

Fall 2025

1. A 25-year-old man comes to the physician because of a 10-year history of progressive weakness of the upper and lower limbs. His 50-year-old maternal uncle has similar symptoms and is wheelchair-bound. Physical examination shows proximal muscle weakness in both upper and lower limbs, mild calf hypertrophy, and a waddling gait. Laboratory studies show mildly elevated serum creatine kinase. Genetic testing is performed. Which of the following results best explains the clinical features in this patient?

- | | |
|---|-----|
| A. Single base pair change, missense mutation | 20% |
| B. Small insertion in a gene causing a frameshift | 20% |
| C. Deletion of 3 exons causing a frameshift | 20% |
| D. In-frame deletion of 7 exons | 20% |
| E. Skewed X-inactivation | 20% |

1. A 25-year-old man comes to the physician because of a 10-year history of progressive weakness of the upper and lower limbs. His 50-year-old maternal uncle has similar symptoms and is wheelchair-bound. Physical examination shows proximal muscle weakness in both upper and lower limbs, mild calf hypertrophy, and a waddling gait. Laboratory studies show mildly elevated serum creatine kinase. Genetic testing is performed. Which of the following results best explains the clinical features in this patient?

- A. Single base pair change, missense mutation 20%
- B. Small insertion in a gene causing a frameshift 20%
- C. Deletion of 3 exons causing a frameshift 20%
- ✓ D. In-frame deletion of 7 exons 20%
- E. Skewed X-inactivation 20%

2. A 16-year-old girl is brought to the physician by her parents for a routine health maintenance examination. She has a history of two fractures following minor trauma, one at age 10 and the other at age 14. Physical examination is unremarkable except for slightly bluish sclera. Genetic testing is performed, and a heterozygous mutation is found in a gene that leads to reduced production of type I collagen. Which of the following molecular mechanisms best describes the cause of this patient's condition?

- | | |
|------------------------------------|-----|
| A. Dominant negative effect | 17% |
| B. Gain-of-function mutation | 17% |
| C. Compound heterozygosity | 17% |
| D. Somatic mosaicism | 17% |
| E. Haploinsufficiency | 17% |
| F. Autosomal recessive inheritance | 17% |

2. A 16-year-old girl is brought to the physician by her parents for a routine health maintenance examination. She has a history of two fractures following minor trauma, one at age 10 and the other at age 14. Physical examination is unremarkable except for slightly bluish sclera. Genetic testing is performed, and a heterozygous mutation is found in a gene that leads to reduced production of type I collagen. Which of the following molecular mechanisms best describes the cause of this patient's condition?

- A. Dominant negative effect 17%
- B. Gain-of-function mutation 17%
- C. Compound heterozygosity 17%
- D. Somatic mosaicism 17%
- ✓ E. Haploinsufficiency 17%
- F. Autosomal recessive inheritance 17%

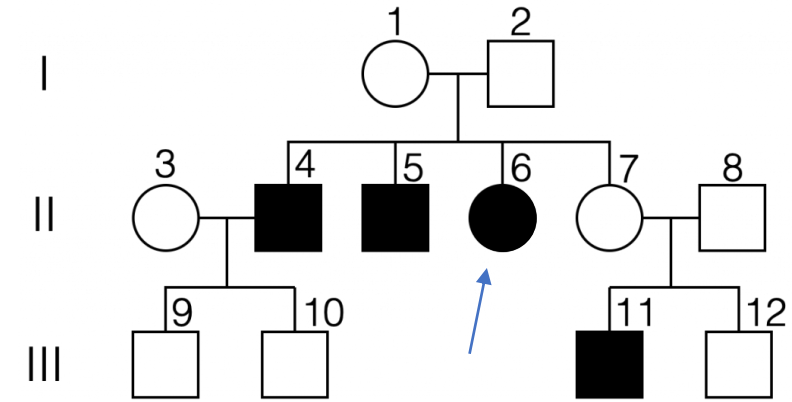
3. A group of investigators is studying the pathogenesis of various musculoskeletal disorders. They are interested in disorders caused by mutations that result in the production of abnormal proteins, which interfere with the function of normal proteins produced by the unaffected allele. Which of the following disorders best illustrates this genetic mechanism?

- A. Becker muscular dystrophy
- B. Osteogenesis imperfecta, perinatal lethal
- C. Achondroplasia
- D. Marfan syndrome
- E. Duchenne muscular dystrophy
- F. Osteogenesis imperfecta, non-deforming

3. A group of investigators is studying the pathogenesis of various musculoskeletal disorders. They are interested in disorders caused by mutations that result in the production of abnormal proteins, which interfere with the function of normal proteins produced by the unaffected allele. Which of the following disorders best illustrates this genetic mechanism?

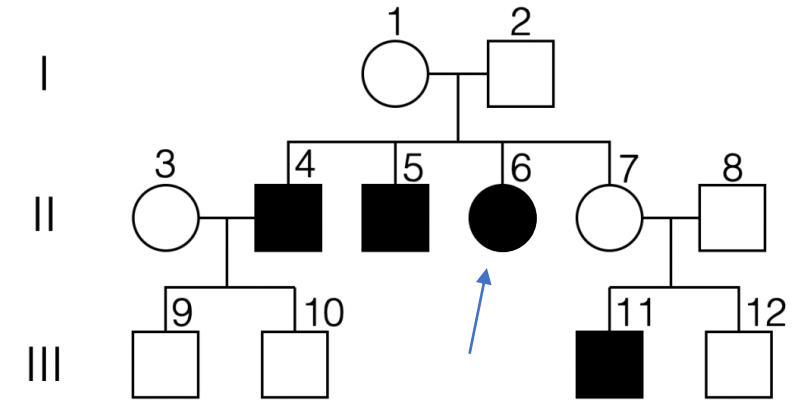
- A. Becker muscular dystrophy
- ✓ B. Osteogenesis imperfecta, perinatal lethal
- C. Achondroplasia
- D. Marfan syndrome
- E. Duchenne muscular dystrophy
- F. Osteogenesis imperfecta, non-deforming

4. A 25-year-old woman comes to the physician because of a 6-year history of progressive fatigue and difficulty climbing stairs. She has 2 older brothers who require mobility assistance, and her nephew has difficulty standing up from the floor. A genogram of the family is shown. Physical examination shows a waddling gait, toe walking, and mild calf hypertrophy. Which of the following genetic mechanisms best describes the milder, later-onset symptoms in this patient compared to her affected brothers and nephew?



- A. X-linked recessive inheritance
- B. Compound heterozygosity
- C. Autosomal recessive inheritance
- D. X-linked dominant inheritance
- E. Skewed X-inactivation

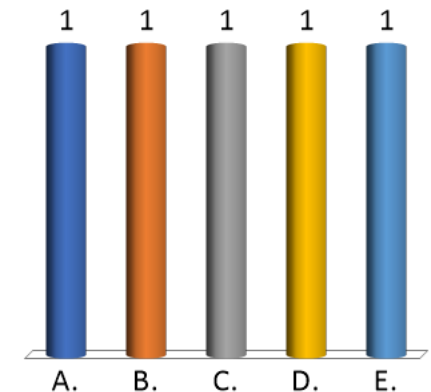
4. A 25-year-old woman comes to the physician because of a 6-year history of progressive fatigue and difficulty climbing stairs. She has 2 older brothers who require mobility assistance, and her nephew has difficulty standing up from the floor. A genogram of the family is shown. Physical examination shows a waddling gait, toe walking, and mild calf hypertrophy. Which of the following genetic mechanisms best describes the milder, later-onset symptoms in this patient compared to her affected brothers and nephew?



- A. X-linked recessive inheritance
- B. Compound heterozygosity
- C. Autosomal recessive inheritance
- D. X-linked dominant inheritance
- ✓ E. Skewed X-inactivation

5. A 31-year-old man comes to the physician because of a 5-month history of progressive muscle weakness. He says the weakness is less in the morning, gradually increases during the day, and worse in the late evening. Vital signs are within normal limits. Physical examination shows ptosis. Serologic studies show autoantibodies to acetylcholine receptors in the postsynaptic membrane of the neuromuscular junction. Which of the following sets of alterations in the neuromuscular junction of the skeletal muscles is most likely in this patient?

Option	Amplitude of End plate potential (EPP)	Amount of acetylcholine released	Cholinesterase concentration in synaptic cleft
A	Increased	Increased	Reduced
B	Reduced	No change	Increase
C	Reduced	Reduced	No change
D	No change	Increased	Increased
E	Reduced	No change	No change



5. A 31-year-old man comes to the physician because of a 5-month history of progressive muscle weakness. He says the weakness is less in the morning, gradually increases during the day, and worse in the late evening. Vital signs are within normal limits. Physical examination shows ptosis. Serologic studies show autoantibodies to acetylcholine receptors in the postsynaptic membrane of the neuromuscular junction. Which of the following sets of alterations in the neuromuscular junction of the skeletal muscles is most likely in this patient?

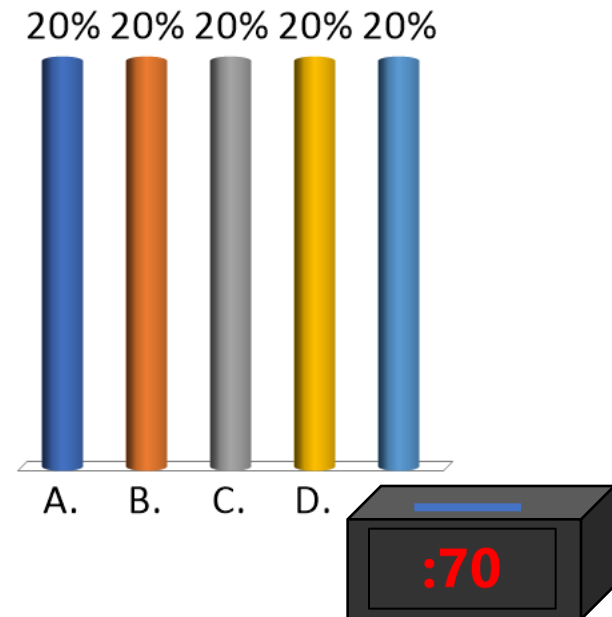
Option	Amplitude of End plate potential (EPP)	Amount of acetylcholine released	Cholinesterase concentration in synaptic cleft
A	Increased	Increased	Reduced
B	Reduced	No change	Increase
C	Reduced	Reduced	No change
D	No change	Increased	Increased
E	Reduced	No change	No change



SOM.MK.II.BPM1.2.MSK.2.PHYS.0060. List the possible sites for blocking neuromuscular transmission in skeletal muscle and provide an example of an agent that could cause blockage at each site.

6. A 53-year-old woman comes to the physician for a follow-up examination. Nine months ago, she was diagnosed with a lung cancer and is currently on chemotherapy. She says she has constipation and muscle weakness in her shoulder, hip and thigh for the past 2 weeks. The muscle weakness improves with repeated use of these muscles. Physical examination shows proximal muscle weakness, dry mouth, and hyporeflexia. Serum studies show autoantibodies. She is diagnosed with Lambert-Eaton myasthenic syndrome. Which of the following mechanisms best explains muscle weakness in this patient?

- A. Overstimulation of the nicotinic ACh receptors
- B. Inhibition of presynaptic voltage-gated calcium channels
- C. Inhibition of calcium binding with the synaptic vesicles
- D. Blocking of the nicotinic ACh receptors by autoantibodies
- E. Inhibition of the acetylcholinesterase enzyme



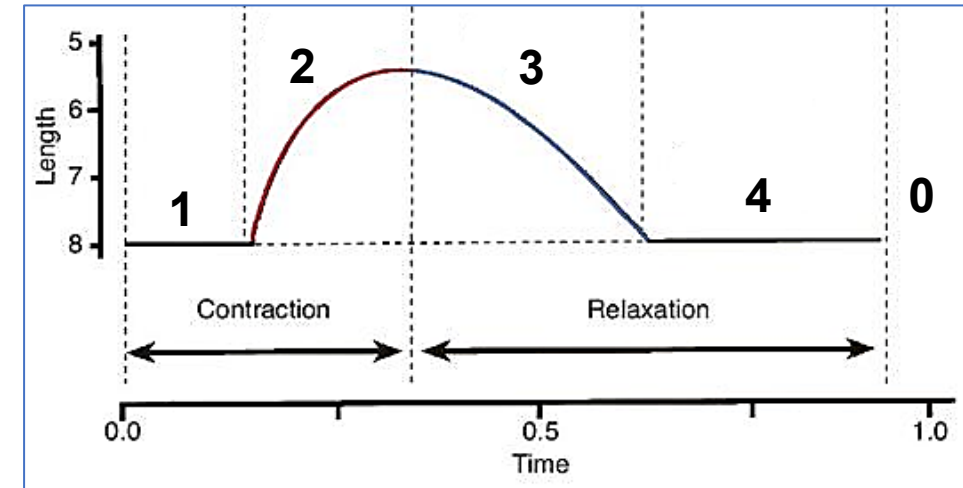
6. A 53-year-old woman comes to the physician for a follow-up examination. Nine months ago, she was diagnosed with a lung cancer and is currently on chemotherapy. She says she has constipation and muscle weakness in her shoulder, hip and thigh for the past 2 weeks. The muscle weakness improves with repeated use of these muscles. Physical examination shows proximal muscle weakness, dry mouth, and hyporeflexia. Serum studies show autoantibodies. She is diagnosed with Lambert-Eaton myasthenic syndrome. Which of the following mechanisms best explains muscle weakness in this patient?

- A. Overstimulation of the nicotinic ACh receptors
- ✓ B. Inhibition of presynaptic voltage-gated calcium channels
- C. Inhibition of calcium binding with the synaptic vesicles
- D. Blocking of the nicotinic ACh receptors by autoantibodies
- E. Inhibition of the acetylcholinesterase enzyme

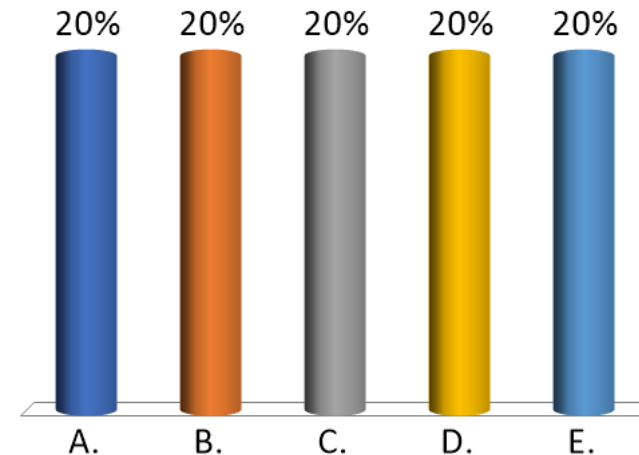
SOM.MK.II.BPM1.2.MSK.2.PHYS.0060. List the possible sites for blocking neuromuscular transmission in skeletal muscle and provide an example of an agent that could cause blockage at each site.



7. An 8-year-old girl is brought to the clinic by her mother because of a 30-hour history of pain in the left arm. The mother says that she pushed and moved a table repeatedly with her arms for about 15 minutes and stopped when she felt pain in the arms. Vital signs are within normal limits. Physical examination shows tenderness in the biceps muscles bilaterally. A graph showing various phases of an isotonic skeletal muscle contraction is shown. During which of the following phases most likely this patient started pushing the table, but unable to move the table?

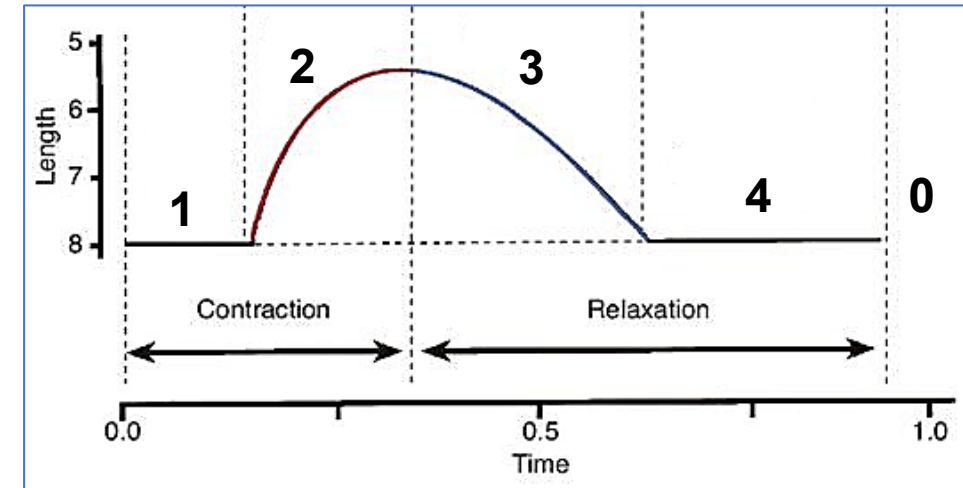


- A. Phase 1
- B. Phase 2
- C. Phase 3
- D. Phase 4
- E. Phase 0

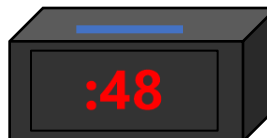
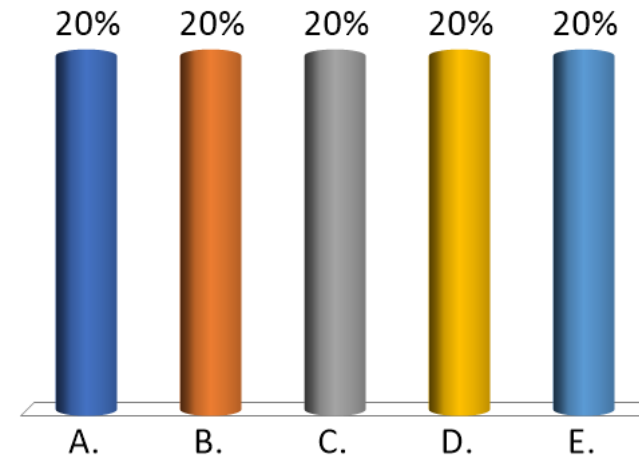


7. An 8-year-old girl is brought to the clinic by her mother because of a 30-hour history of pain in the left arm. The mother says that she pushed and moved a table repeatedly with her arms for about 15 minutes and stopped when she felt pain in the arms. Vital signs are within normal limits. Physical examination shows tenderness in the biceps muscles bilaterally. A graph showing various phases of an isotonic skeletal muscle contraction is shown. During which of the following phases most likely this patient started pushing the table, but unable to move the table?

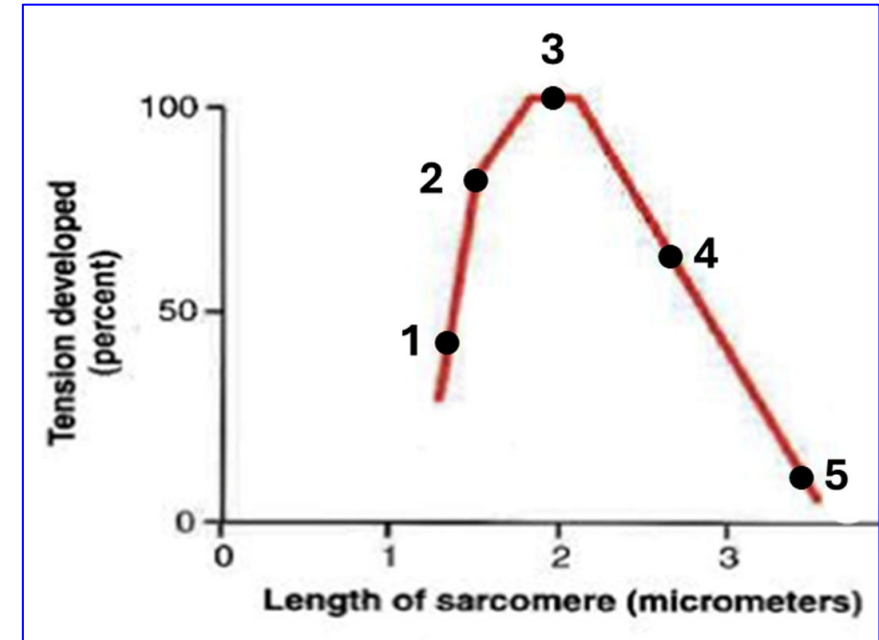
- ✓ A. Phase 1
- B. Phase 2
- C. Phase 3
- D. Phase 4
- E. Phase 0



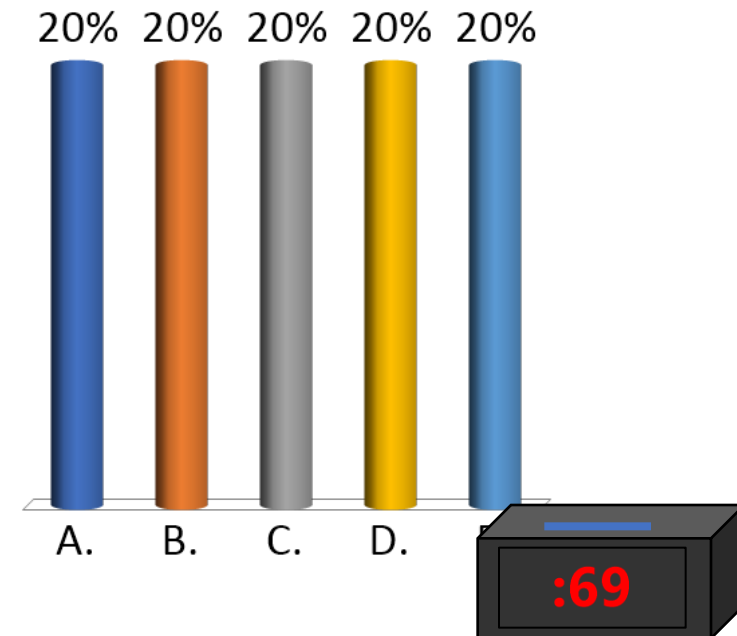
SOM.MK. II. BPM1.2.MSK.2.PHYS.0062. Compare and contrast between an isometric and isotonic contraction.



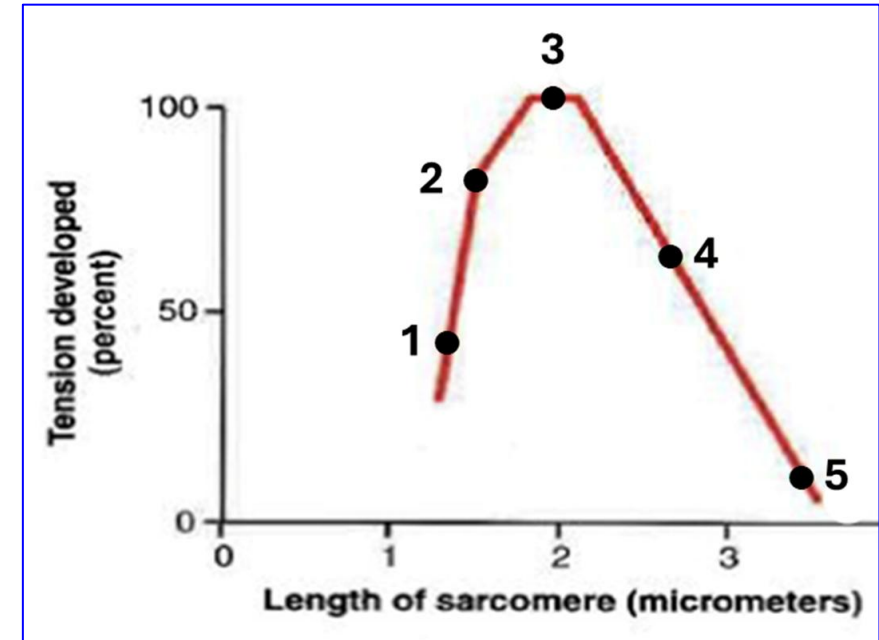
8. Investigators conduct a study that evaluates the effect of altering the resting length of a skeletal muscle on the muscle tension. A length-tension relationship of the skeletal muscle is recorded as shown in the figure. Which of the following statements best explains the muscle tension at point 5 in the figure?



- A. Decreased amount of ATP availability
- B. Increased overlap of actin and myosin filaments
- C. Decreased overlap of actin and myosin filaments
- D. Increased amount of calcium release from the sarcoplasmic reticulum
- E. Least number of cross bridge cycling



8. Investigators conduct a study that evaluates the effect of altering the resting length of a skeletal muscle on the muscle tension. A length-tension relationship of the skeletal muscle is recorded as shown in the figure. Which of the following statements best explains the muscle tension at point 5 in the figure?

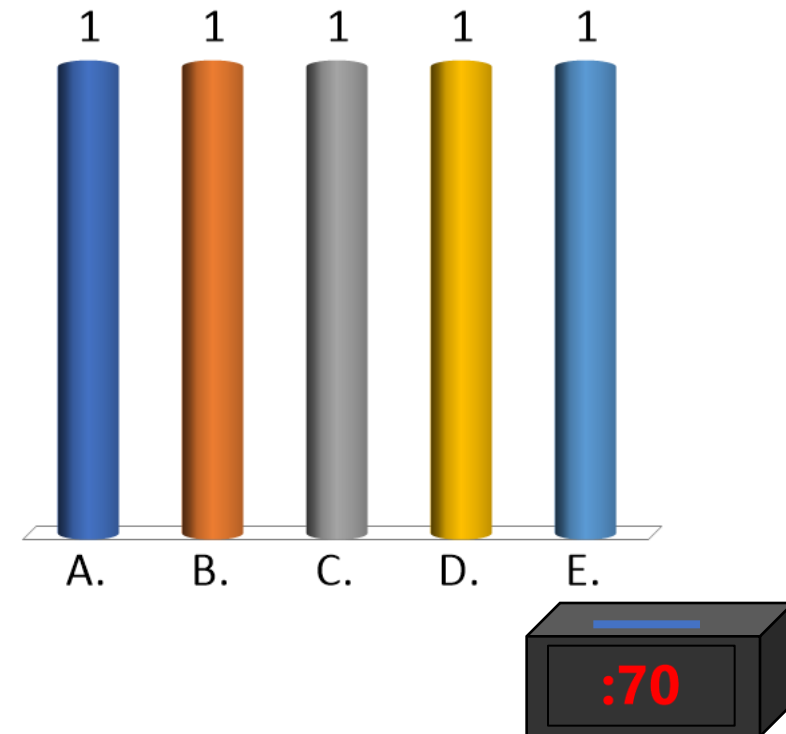


SOM.MK.II.BPM1.2.MSK.2.PHYS.0061 Describe the role of the myosin cross bridges acting in parallel to determine active force.

- A. Decreased amount of ATP availability
- B. Increased overlap of actin and myosin filaments
- C. Decreased overlap of actin and myosin filaments
- D. Increased amount of calcium release from the sarcoplasmic reticulum
- ✓ E. Least number of cross bridge cycling

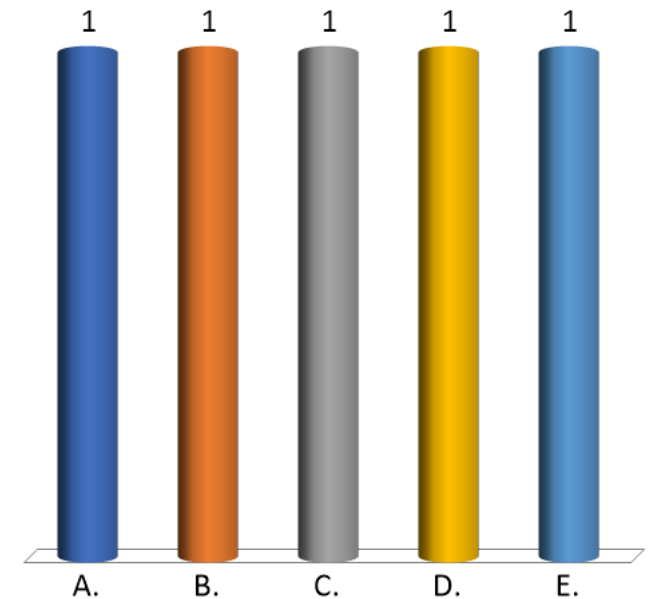
9. A 24-year-old man comes to the physician for a follow-up examination. Ten weeks ago, he sustained an injury to his right leg while playing soccer. He has no pain but has weakness in the right leg. Vital signs are within normal limits. Physical examination of the lower limbs show no abnormalities. The physician tests the strength of his right leg muscles by asking him to lift graded weights starting from 1-kg to 10-kg. He can lift lighter weights but is unsuccessful in lifting 7-kg weight. Which of the following mechanisms best explains the patient's inability to lift this weight?

- A. The load has overstretched the muscle
- B. The load is unable to produce a tetanic contraction of the muscles
- C. The load is greater than the tension developed in the muscles
- D. The load and the muscle tension are exactly equal
- E. The tension developed in the muscles is greater than the load

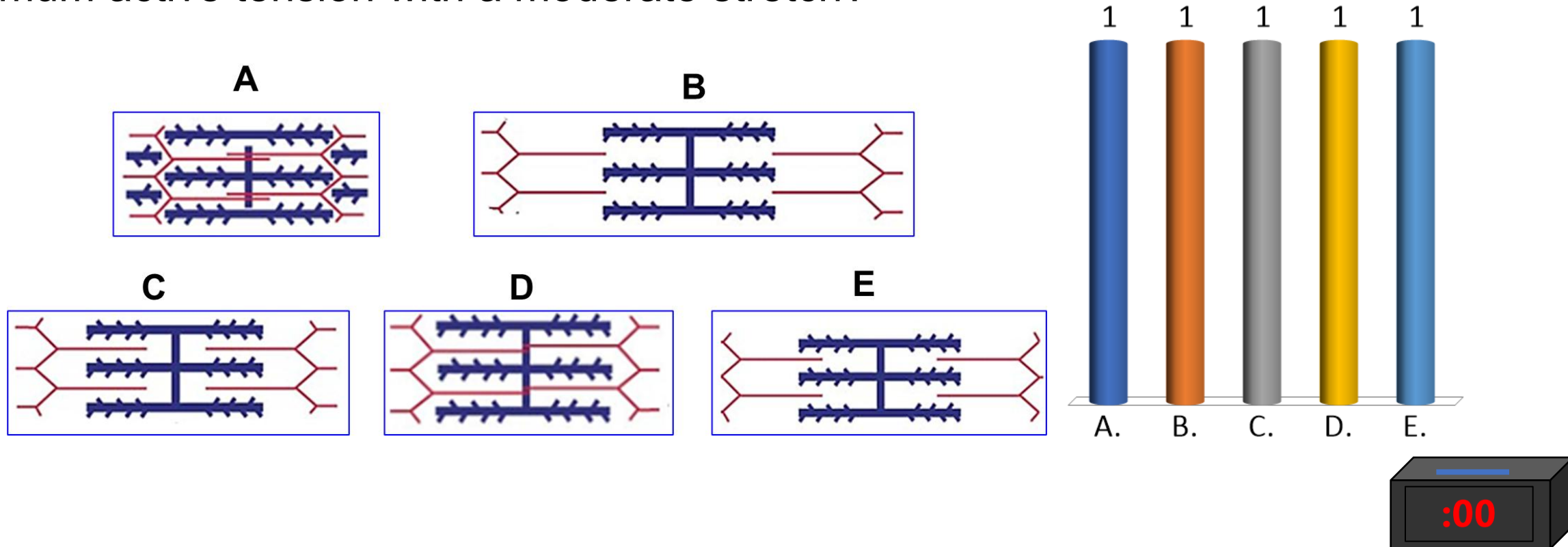


9. A 24-year-old man comes to the physician for a follow-up examination. Ten weeks ago, he sustained an injury to his right leg while playing soccer. He has no pain but has weakness in the right leg. Vital signs are within normal limits. Physical examination of the lower limbs show no abnormalities. The physician tests the strength of his right leg muscles by asking him to lift graded weights starting from 1-kg to 10-kg. He can lift lighter weights but is unsuccessful in lifting 7-kg weight. Which of the following mechanisms best explains the patient's inability to lift this weight?

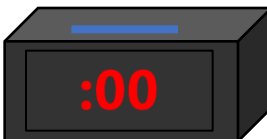
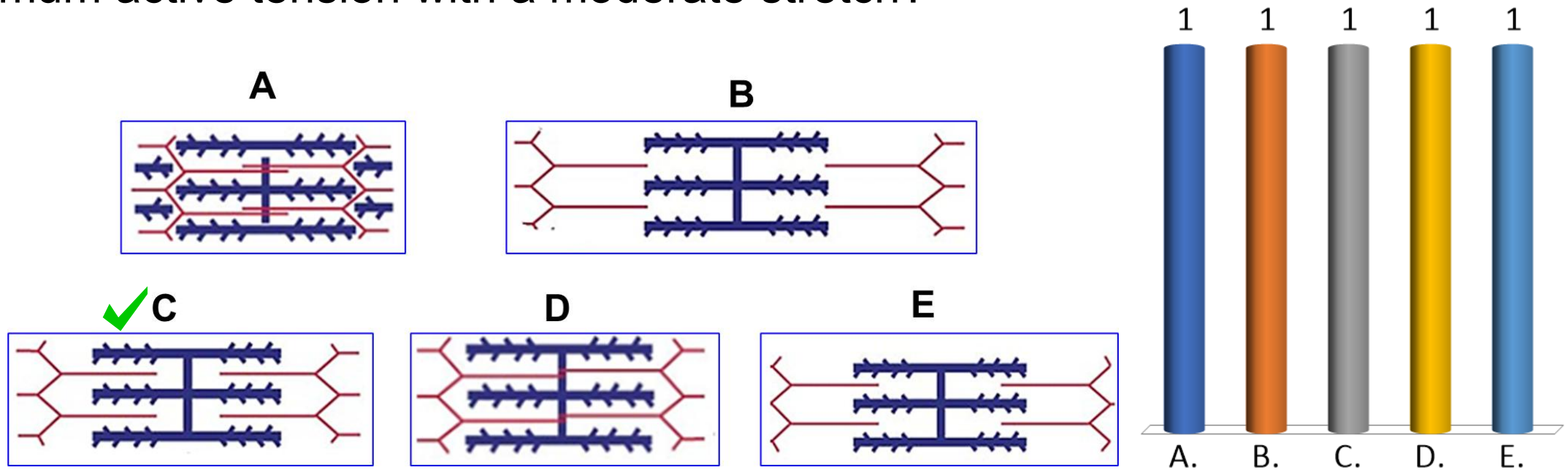
- A. The load has overstretched the muscle
- B. The load is unable to produce a tetanic contraction of the muscles
- ✓ C. The load is greater than the tension developed in the muscles
- D. The load and the muscle tension are exactly equal
- E. The tension developed in the muscles is greater than the load



10. Investigators conduct a study that relates the length of a skeletal muscle to its active tension. A freshly dissected rat's leg muscle with intact motor nerve is used for this study. Series of muscle tensions are recorded by altering the length of the muscle by passive stretch. It is observed that the active tension increased by stretching the muscles initially but started declining as the muscle is over-stretched. Which of the following figures best represents the sarcomere that produces the maximum active tension with a moderate stretch?

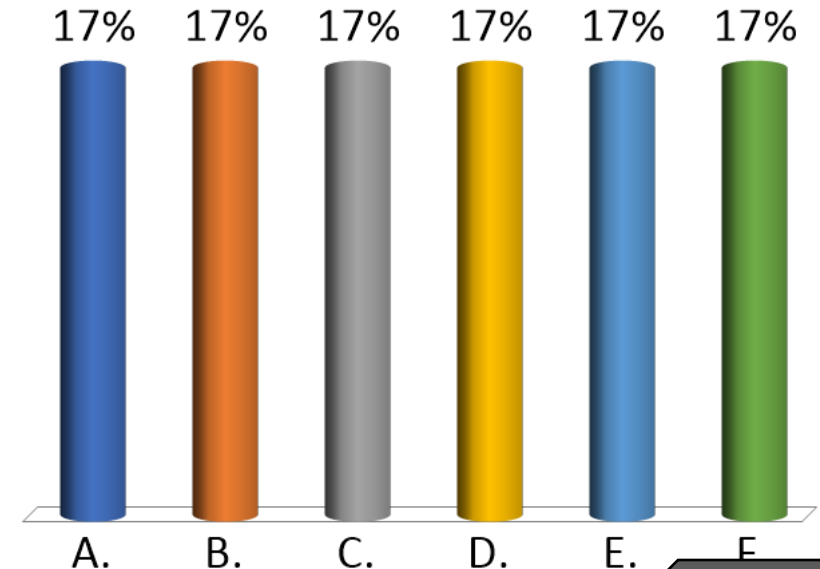


10. Investigators conduct a study that relates the length of a skeletal muscle to its active tension. A freshly dissected rat's leg muscle with intact motor nerve is used for this study. Series of muscle tensions are recorded by altering the length of the muscle by passive stretch. It is observed that the active tension increased by stretching the muscles initially but started declining as the muscle is over-stretched. Which of the following figures best represents the sarcomere that produces the maximum active tension with a moderate stretch?



11. A 26-year-old, healthy man volunteers for a study related to the mechanics of skeletal muscle contraction. His brachioradialis muscle in the right forearm is selected for the study. The branch of the radial nerve that supplies to the muscle is isolated. Muscle contraction in response to an electrical stimulation of this nerve is recorded. A single stimulus produces a muscle twitch. Repetitive stimulation of the nerve generates an increased muscle tension, and the muscle remains in sustained contraction. Which of the following mechanisms best explains the finding with repetitive stimulation of the muscle?

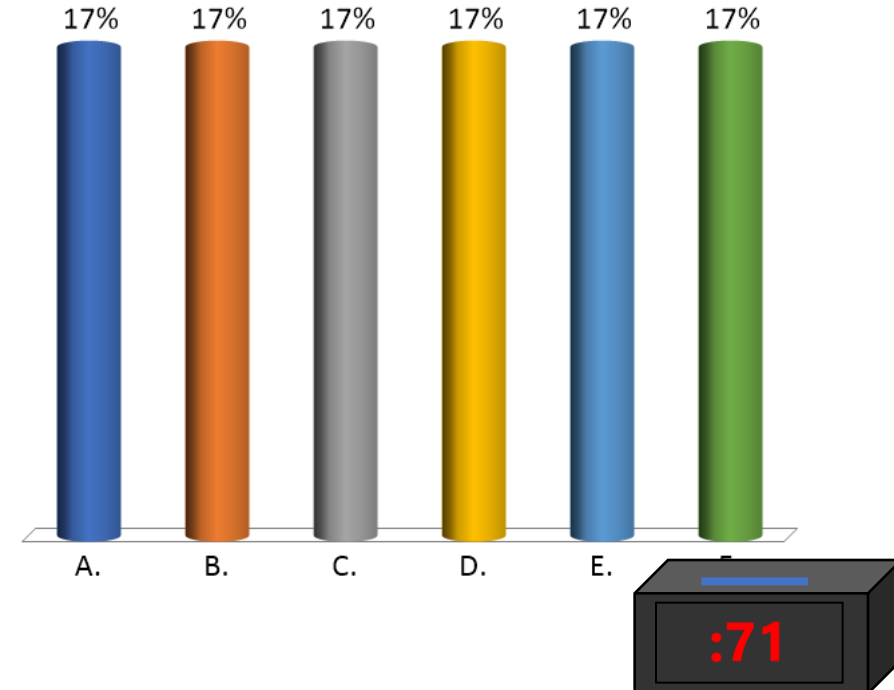
- A. Spatial summation
- B. Motor unit recruitment
- C. Muscle fatigue
- D. Temporal summation
- E. Increased rate of cross bridge cycling
- F. Macromotor unit firing



11. A 26-year-old, healthy man volunteers for a study related to the mechanics of skeletal muscle contraction. His brachioradialis muscle in the right forearm is selected for the study. The branch of the radial nerve that supplies to the muscle is isolated. Muscle contraction in response to an electrical stimulation of this nerve is recorded. A single stimulus produces a muscle twitch. Repetitive stimulation of the nerve generates an increased muscle tension, and the muscle remains in sustained contraction. Which of the following mechanisms best explains the finding with repetitive stimulation of the muscle?

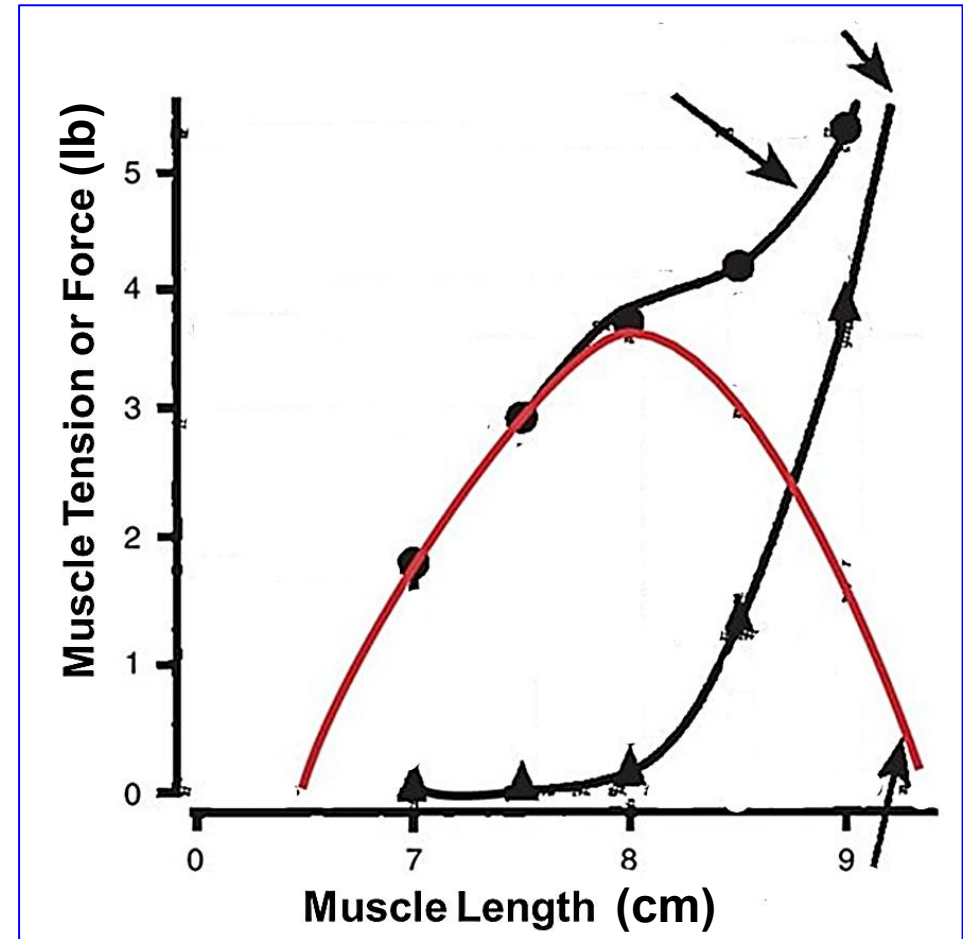
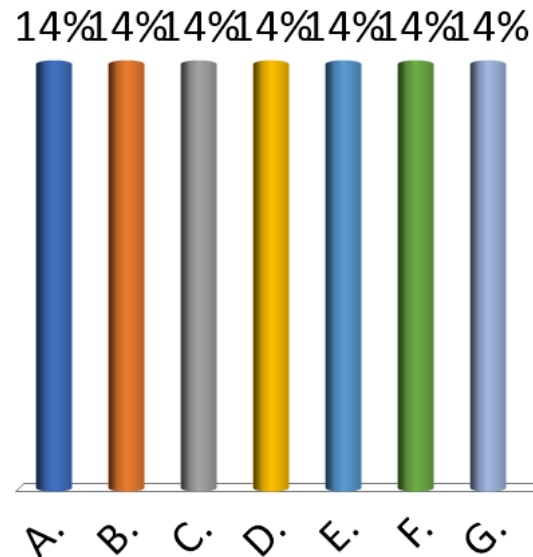
- A. Spatial summation
- B. Motor unit recruitment
- C. Muscle fatigue
- ✓ D. Temporal summation
- E. Increased rate of cross bridge cycling
- F. Macromotor unit firing

SOM.MK.II.BPM1.2.MSK.2.PHYS.0065. Distinguish between a twitch and tetanus in skeletal muscle and explain why a twitch is smaller in amplitude than tetanus and the continuum of force development between a twitch and tetanus including the intracellular events.



12. Investigators conduct a study that evaluates the effect of a preload on muscle tensions in a skeletal muscle. A length-tension relationship of the whole skeletal muscle is shown in the graph. The arrows in the graph indicate the total tension, the active tension, and the passive tension. Which of the following values best represents the active tension when the preload stretches the muscle to 9-cm muscle length?

- A. 0 (zero) lb
- B. 0.8 lb
- C. 1.7 lb
- D. 2.4 lb
- E. 3.0 lb
- F. 3.7 lb
- G. 5.4 lb



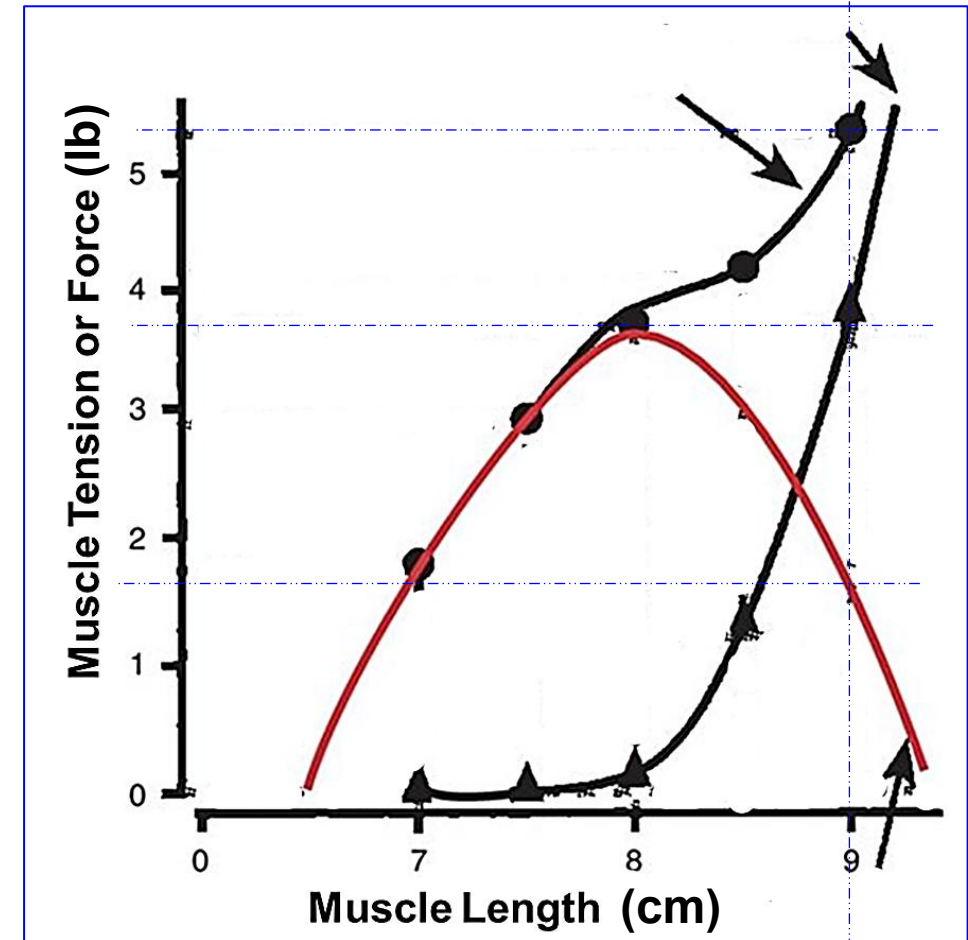
$$TT = PT + AT$$

$$5.4 = 3.7 + 1.7$$

12. Investigators conduct a study that evaluates the effect of a preload on muscle tensions in a skeletal muscle. A length-tension relationship of the whole skeletal muscle is shown in the graph. The arrows in the graph indicate the total tension, the active tension, and the passive tension. Which of the following values best represents the active tension when the preload stretches the muscle to 9-cm muscle length?

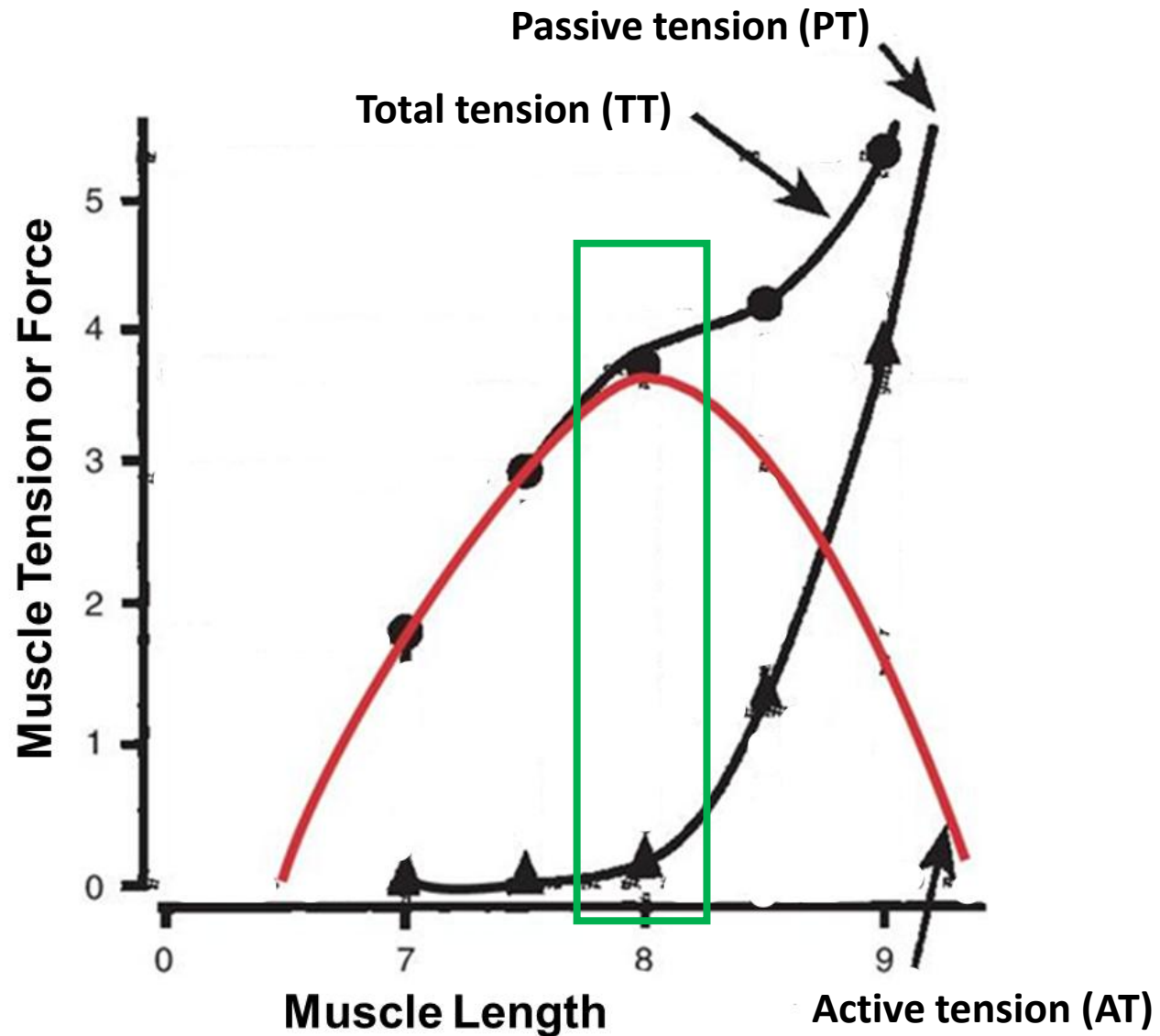
- A. 0 (zero) lb
- B. 0.8 lb
- ✓ C. 1.7 lb
- D. 2.4 lb
- E. 3.0 lb
- F. 3.7 lb
- G. 5.4 lb

SOM.MK. II. BPM1.2.MSK.2.PHYS.0063. Illustrate the length versus force diagram for muscle and label the three lines that represent passive (resting), active, and total force. Explain the molecular origin of these forces in the three muscle types.



$$\begin{aligned} TT &= PT + AT \\ 5.4 &= 3.7 + 1.7 \end{aligned}$$

Length-Tension Relationship in the whole intact muscle

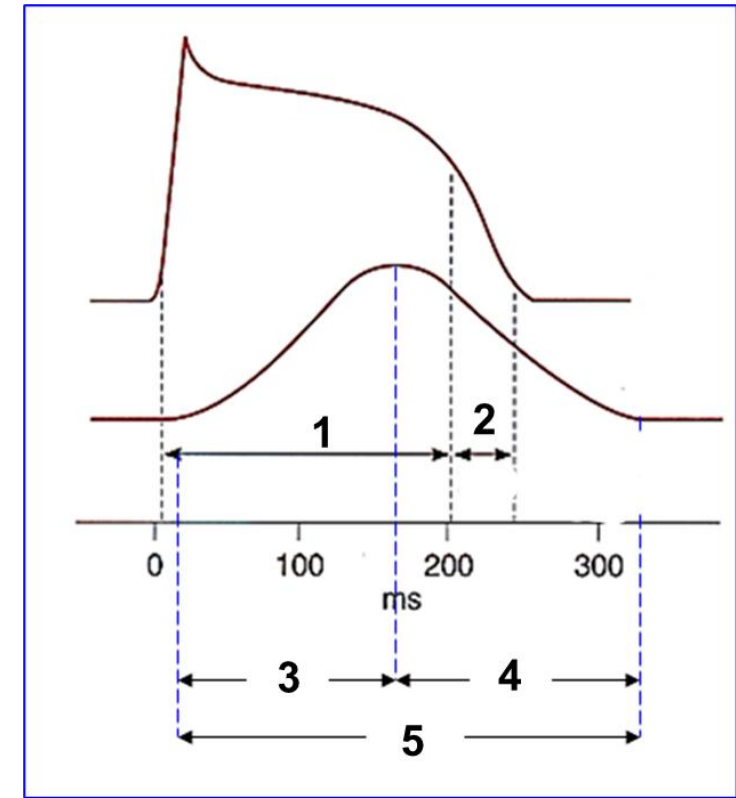


At any given time,
Total tension (TT) is equal to
sum of Passive and Active Tensions.

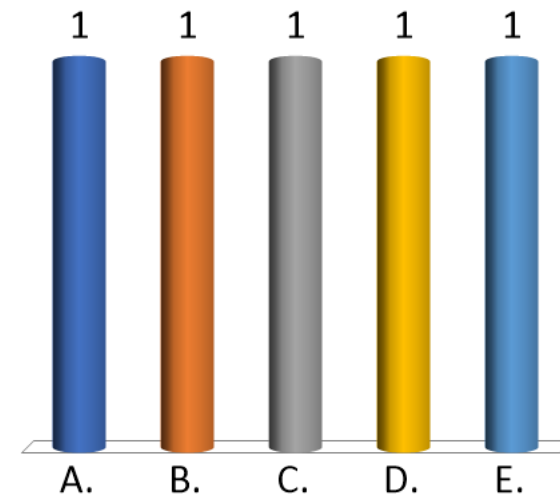
$$TT = PT + AT$$

The operational range of a skeletal muscle is shown in the green box. In this range, the muscle has optimal length (L_0) and produces maximum active tension.

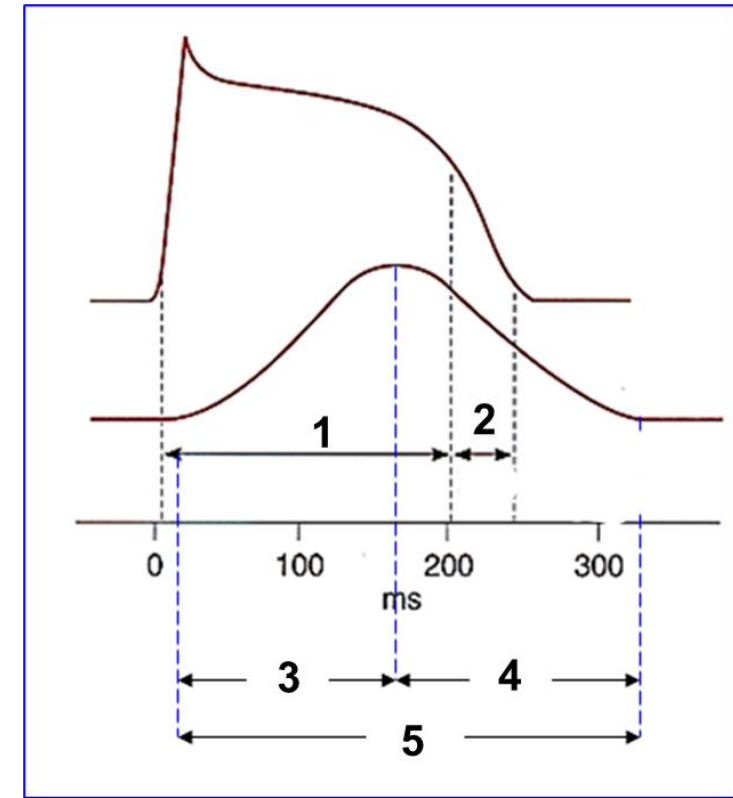
13. Investigators conduct a study that relates the electrical and mechanical properties of cardiac muscle using a chipmunk's heart in vivo. The heart is denervated and stimulated directly by an electrode. It is observed that increasing the frequency of stimulation does not make the ventricle undergo tetanic contraction. The recording of electrical and mechanical events to a single stimulus is shown in the graph. Which of the following labels in the graph most likely represents the absolute refractory period in this cardiac muscle?



- A. Label 1
- B. Label 2
- C. Label 3
- D. Label 4
- E. Label 5



13. Investigators conduct a study that relates the electrical and mechanical properties of cardiac muscle using a chipmunk's heart in vivo. The heart is denervated and stimulated directly by an electrode. It is observed that increasing the frequency of stimulation does not make the ventricle undergo tetanic contraction. The recording of electrical and mechanical events to a single stimulus is shown in the graph. Which of the following labels in the graph most likely represents the absolute refractory period in this cardiac muscle?

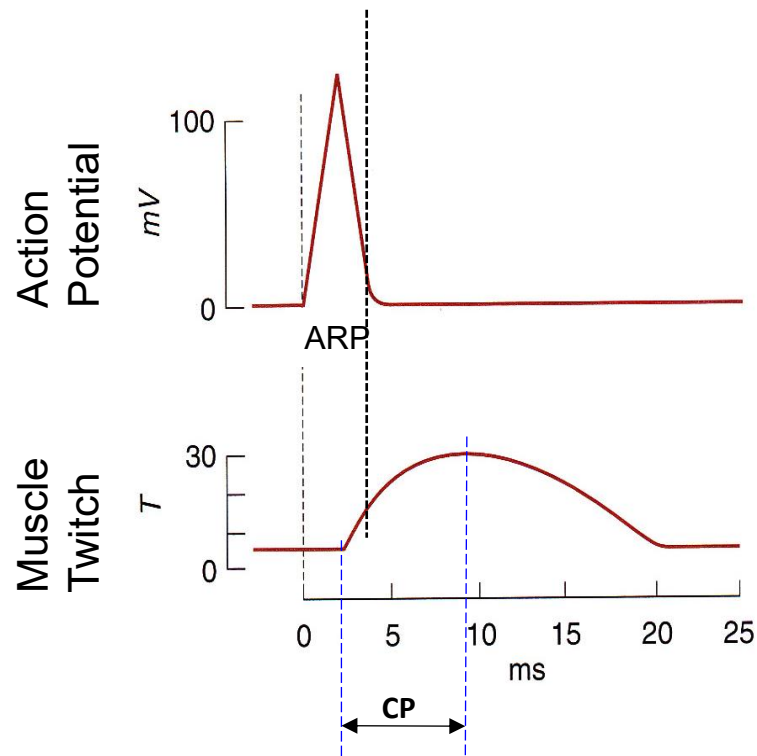


- ✓ A. Label 1
- B. Label 2
- C. Label 3
- D. Label 4
- E. Label 5

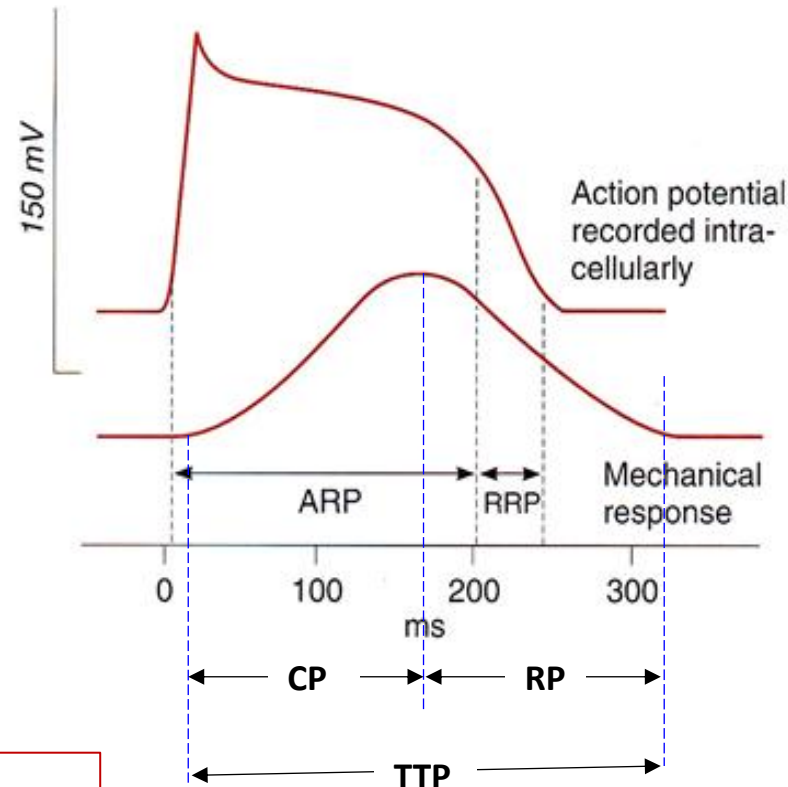
SOM.MK.II.BPM1.2.MSK.2.PHYS.0067 Contrast the duration of the action potential and the refractory period in a cardiac muscle, a skeletal muscle, and a nerve. Sketch the temporal relationship between an action potential in a cardiac muscle cell and the resulting contraction (twitch) of that cell. On the basis of that graph, explain why cardiac muscle cannot remain in a state of sustained (tetanic) contraction.

Comparison of electrical (Action potential) and mechanical (contraction) events in skeletal and cardiac muscles

Skeletal



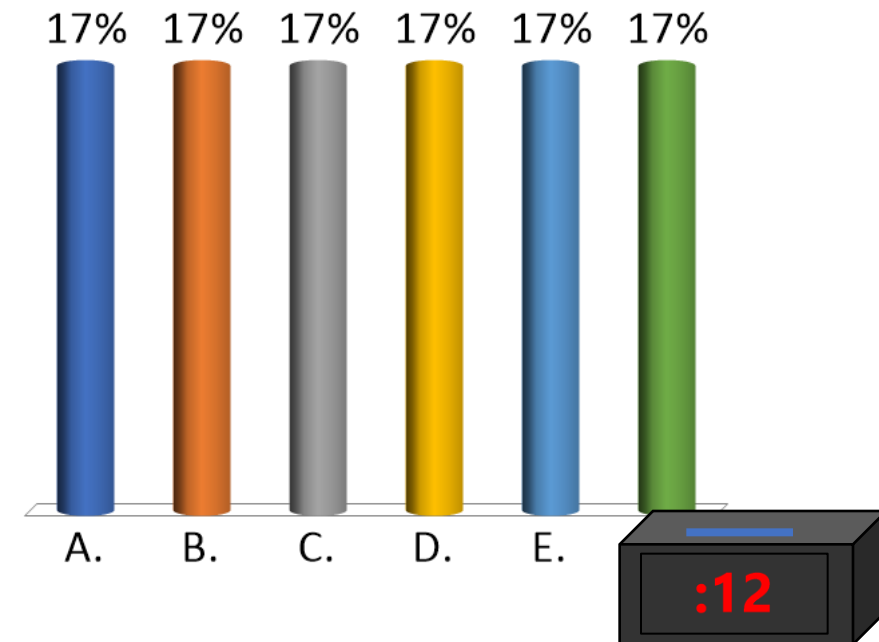
Cardiac



**CP=contraction period; RP=Relaxation period;
TTP=Total twitch period**

14. A 67-year-old woman comes to the physician because of a 4-day history of shortness of breath at nights and feet swelling after prolonged sitting. She has a 10-year history of heart failure and is not adherent to medications. Her pulse is 86/min, respirations are 22/min, and blood pressure is 132/82 mm Hg. Physical examination shows distended neck veins, tender hepatomegaly, and pitting edema of the feet. Coarse crackles are heard over the lower lung fields bilaterally. She is prescribed milrinone, a positive inotropic agent. Her condition improves after this treatment. Which of the following mechanisms best explains increased myocardial calcium concentration in this patient after treatment?

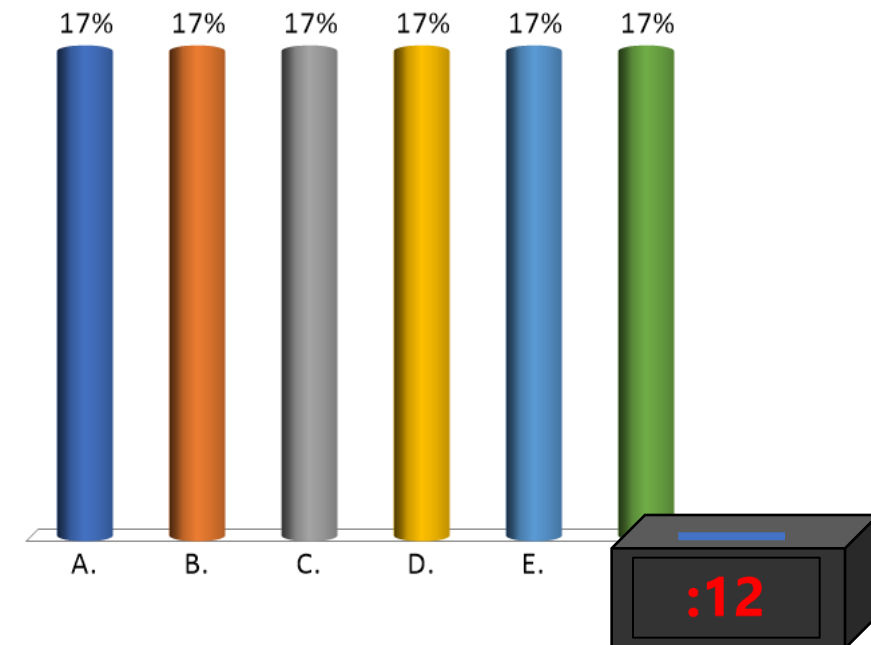
- A. Increased breakdown of cAMP
- B. Reduction in intracellular cAMP concentration
- C. Stimulation of beta-1 receptors
- D. Reduced calcium efflux
- E. Reduced breakdown of cAMP
- F. Reduced calcium influx



14. A 67-year-old woman comes to the physician because of a 4-day history of shortness of breath at nights and feet swelling after prolonged sitting. She has a 10-year history of heart failure and is not adherent to medications. Her pulse is 86/min, respirations are 22/min, and blood pressure is 132/82 mm Hg. Physical examination shows distended neck veins, tender hepatomegaly, and pitting edema of the feet. Coarse crackles are heard over the lower lung fields bilaterally. She is prescribed milrinone, a positive inotropic agent. Her condition improves after this treatment. Which of the following mechanisms best explains increased myocardial calcium concentration in this patient after treatment?

- A. Increased breakdown of cAMP
- B. Reduction in intracellular cAMP concentration
- C. Stimulation of beta-1 receptors
- D. Reduced calcium efflux
- ✓ E. Reduced breakdown of cAMP
- F. Reduced calcium influx

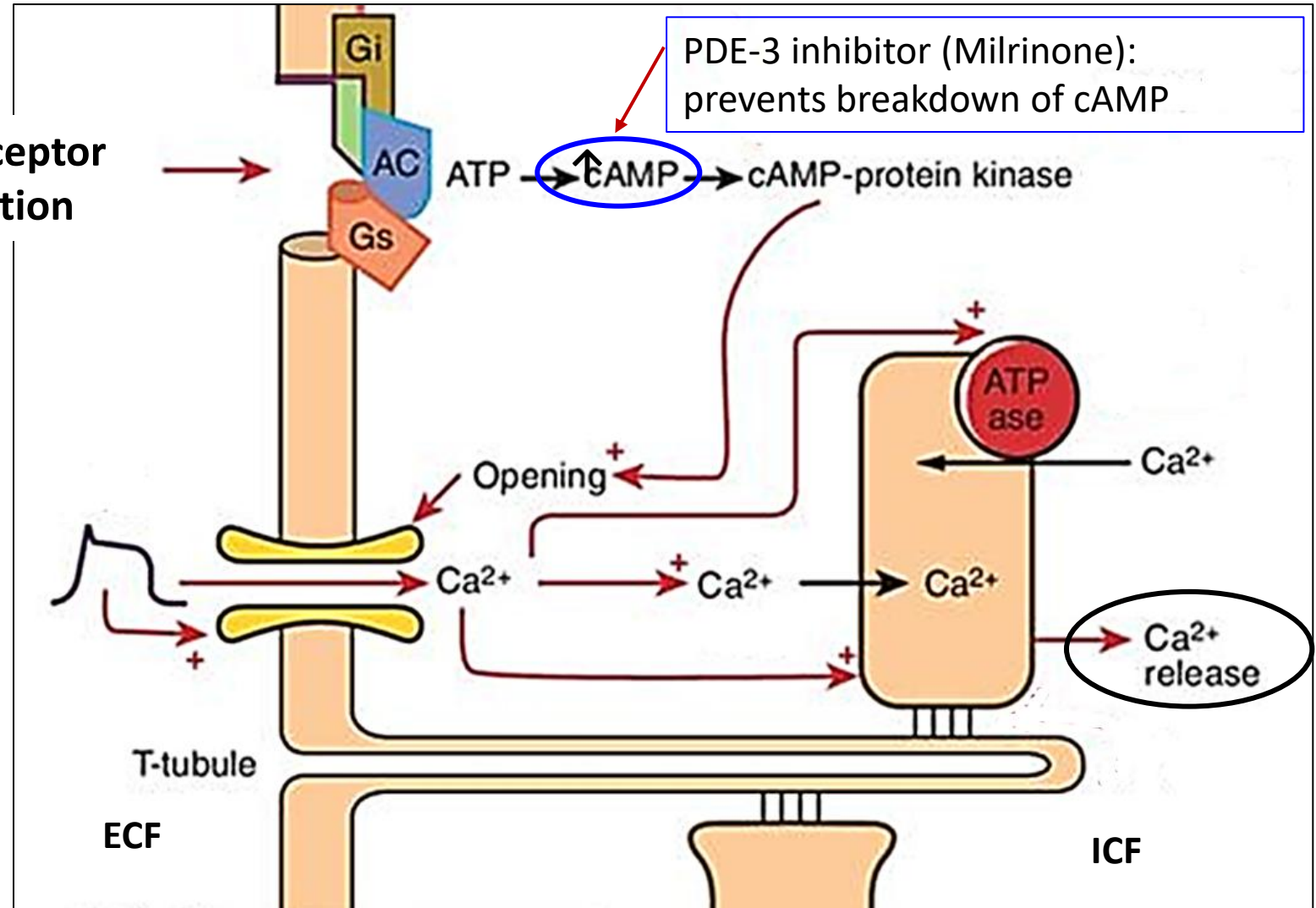
SOM.MK.II.BPM1.2.MSK.2.PHYS.0071 . Identify the role of extracellular calcium in cardiac muscle contraction. Identify other sources of calcium that mediate excitation-contraction coupling and describe how intracellular calcium concentration modulates the strength of cardiac muscle contraction (Inotropism).



1. Increasing Ca^{2+} entry occurs by increasing cAMP inside the myocardium

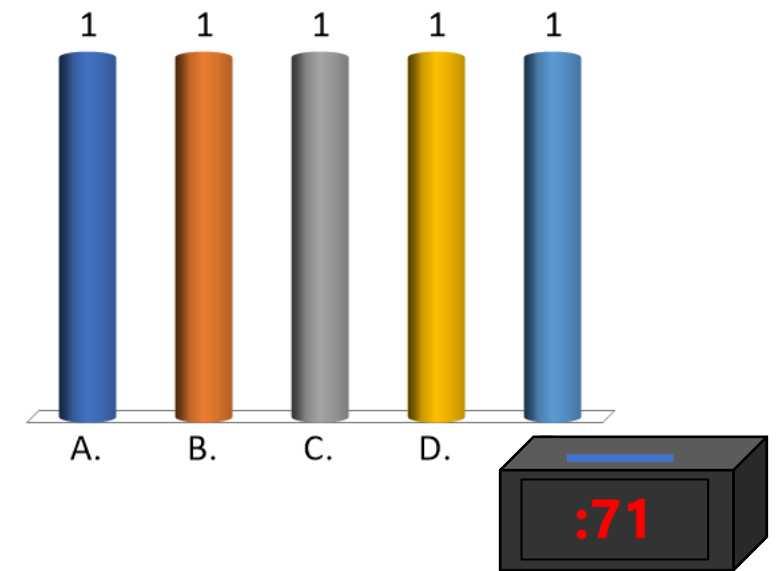
Catecholamines,
Sympathetic stimulation,
Beta-receptor agonists,
Exercise

Beta1 receptor
stimulation



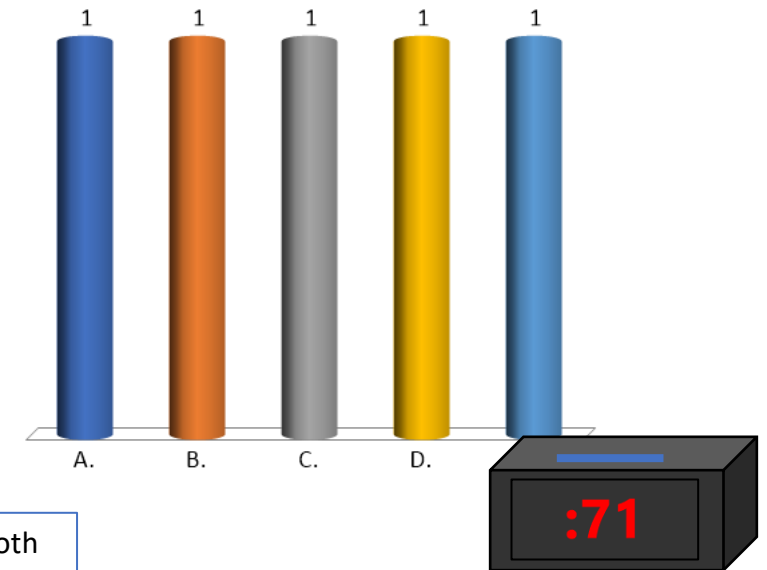
15. A 27-year-old man is brought to the emergency department by his wife because of a 2-hour history of chest pain, agitation, restlessness, and headache. He has a 2-year history of substance use disorder. His pulse is 124/min and blood pressure is 164/98 mm Hg. Physical examination shows dilated pupils and diaphoresis. Mental status examination shows psychosis and impaired judgement. He is diagnosed with acute cocaine intoxication. His blood pressure is controlled by sodium nitroprusside that increases nitric oxide in the smooth muscles of the blood vessels. Which of the following mechanisms best explains relaxation of the vascular smooth muscles in this patient?

- A. Increased myosin light chain kinase activity
- B. Decreased myosin light chain phosphatase activity
- C. Increased intracellular cAMP concentration
- D. Decreased myosin light chain kinase activity
- E. Decreased calcium-ATPase activity

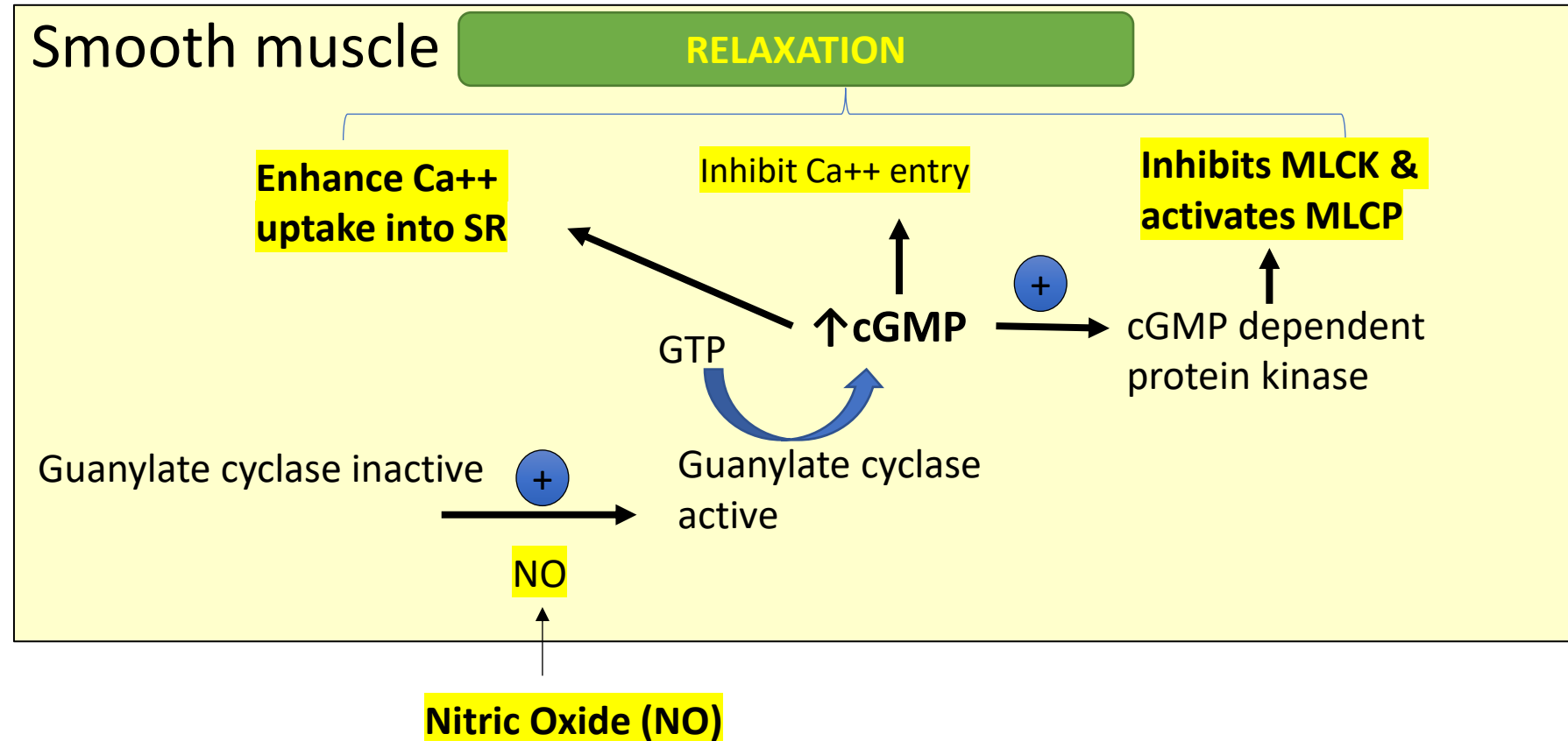


15. A 27-year-old man is brought to the emergency department by his wife because of a 2-hour history of chest pain, agitation, restlessness, and headache. He has a 2-year history of substance use disorder. His pulse is 124/min and blood pressure is 164/98 mm Hg. Physical examination shows dilated pupils and diaphoresis. Mental status examination shows psychosis and impaired judgement. He is diagnosed with acute cocaine intoxication. His blood pressure is controlled by sodium nitroprusside that increases nitric oxide in the smooth muscles of the blood vessels. Which of the following mechanisms best explains relaxation of the vascular smooth muscles in this patient?

- A. Increased myosin light chain kinase activity
- B. Decreased myosin light chain phosphatase activity
- C. Increased intracellular cAMP concentration
- ✓ D. Decreased myosin light chain kinase activity
- E. Decreased calcium-ATPase activity



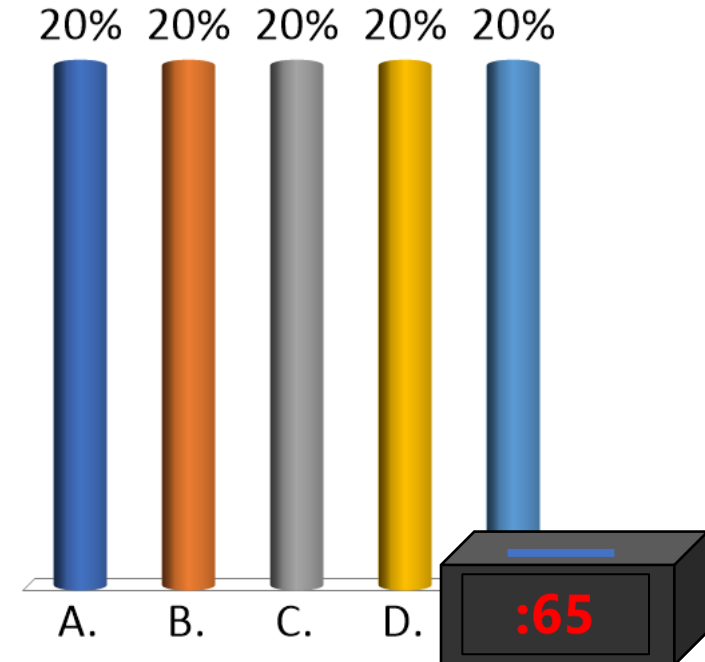
Nitric Oxide Relaxes Smooth muscles: cGMP pathway



- Nitrates used in angina increase NO in vascular smooth muscles improve blood flow to heart.

16. Investigators conduct a study that evaluates the effect of a new drug on the contraction of intestinal smooth muscles. Investigators inject graded doses of the drug and record force of contraction. They observe that the drug increases the release of acetylcholine which binds with the muscarinic receptors in the intestinal smooth muscles. Which of the following mechanisms best explains the action of this drug on the contraction of intestinal smooth muscles?

- A. Depolarization of cells by electromechanical coupling
- B. Inhibition of myosin light chain kinase activity
- C. Activation of Gq-protein and increase in IP3 concentration in the smooth muscles
- D. Activation of adenylyl cyclase and increase in cAMP concentration in the smooth muscles
- E. Activation of voltage-gated calcium channels and influx of calcium



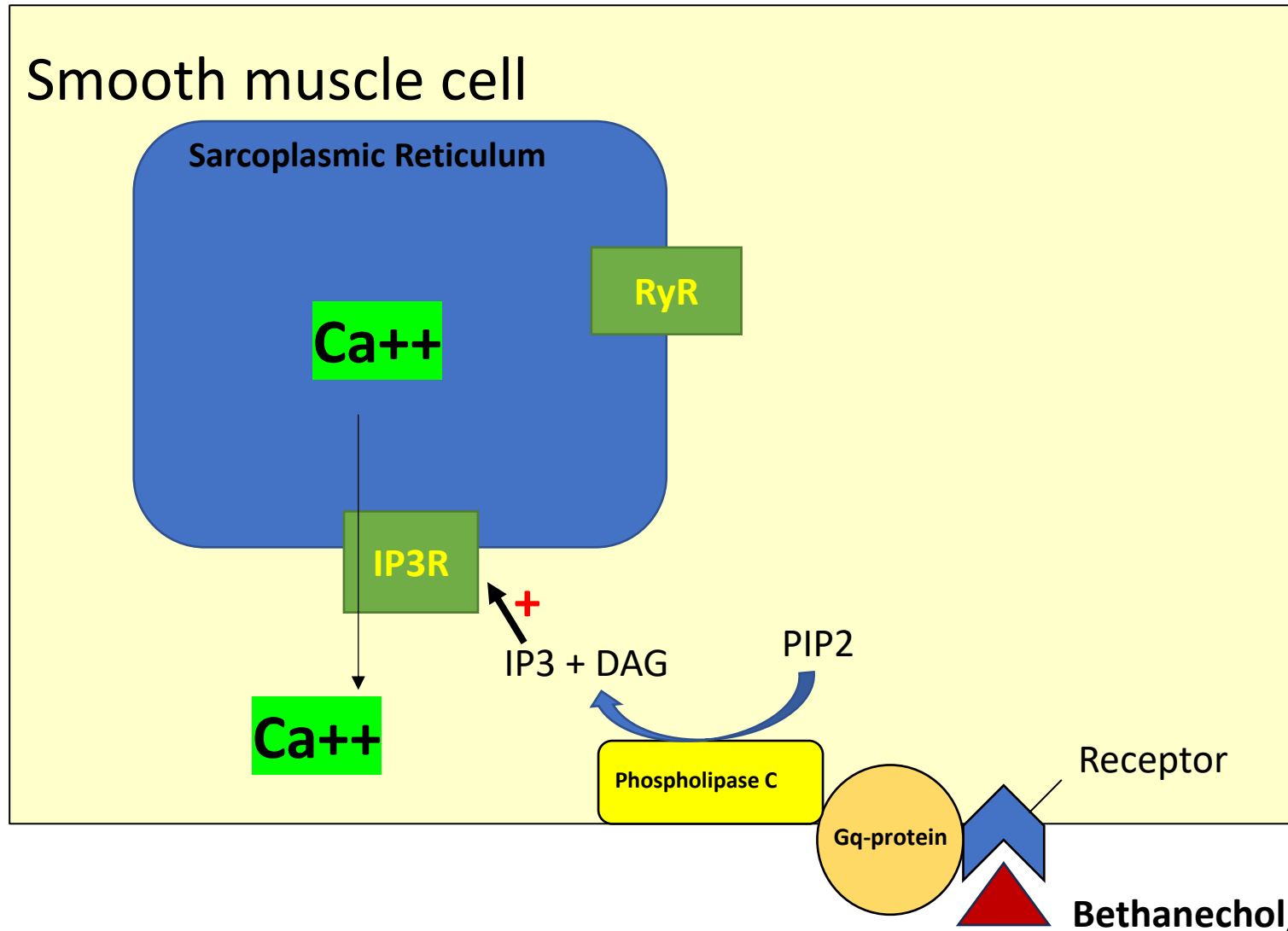
16. Investigators conduct a study that evaluates the effect of a new drug on the contraction of intestinal smooth muscles. Investigators inject graded doses of the drug and record force of contraction. They observe that the drug increases the release of acetylcholine which binds with the muscarinic receptors in the intestinal smooth muscles. Which of the following mechanisms best explains the action of this drug on the contraction of intestinal smooth muscles?

- A. Depolarization of cells by electromechanical coupling
- B. Inhibition of myosin light chain kinase activity
- ✓ C. Activation of Gq-protein and increase in IP3 concentration in the smooth muscles
- D. Activation of adenylyl cyclase and increase in cAMP concentration in the smooth muscles
- E. Activation of voltage-gated calcium channels and influx of calcium

SOM.MK.II.BPM1.2.MSK.2.PHYS.0074. List the intracellular pathways that control the contraction and relaxation in smooth muscle cells and explain the differences between electromechanical and pharmacomechanical coupling



IP3 pathway: Examples



- Norepinephrine, Epinephrine & Dopamine (via alpha adrenoreceptors) in vascular sm muscles
- Acetylcholine (via M receptors) in GIT
- Angiotensin II (via angiotensin II receptors) in vascular sm muscles
- Endothelin (via ET_A receptors)

Bethanechol, Ach-through muscarinic receptors

You've worked hard for this
exam, so give your best effort!

**Best Wishes
To You.**

